Open problems:
Superpositions correspond to multiple decompositions

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

No item below blocks the verification of a main claim of the brief.

From the brief

1. **No classical superpositions.** The introduction argues that "mixtures of different classical states always yield a different ensemble", so that classical superpositions are not possible. The uniqueness of the decomposition of a classical ensemble (a probability measure) into pure (Dirac) states, and the resulting failure of the classical analogue of QSUP-MIX, are not formalized. (Brief, Sec. 1.)

2. **Superpositions of more than two states.** The brief treats superpositions of two states. Whether a superposition of n linearly independent states corresponds in the same way to an ensemble with a decomposition into those n states and one into ϕ and some other states is not addressed. (Brief, Sec. 2, QSUP-PUR, QSUP-MIX.)

From the formalization

1. **Density operators.** QSUP-MIX is stated for convex combinations of pure states without the trace; that ρ is a density operator in the sense of QST (positive, trace one, and trace class in infinite dimension) is not checked. (Log, "Differences".)